

991 + 981 , MY 2012 =>

Diagnosis information - Headlights Central

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Diagnostic information - Headlight control unit (central)

Control module faulty

NOTE

- ♦ The lights should always be switched off before removing and installing the headlights.
- ♦ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Control module** Headlights faulty

Fault effects

- ♦ System switches off
- ♦ Display in instrument cluster

000001

Diagnosis information - Headlight control unit (central)

Control module , not coded

Control module , incorrectly coded

NOTE

- ♦ The lights should always be switched off before removing and installing the headlights.
- ♦ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Headlight control unit (central) not coded
- ♦ Headlight control unit (central) incorrectly coded
- ♦ Headlight control unit (central) Faulty

000002

Diagnostic information - valid for all control units

Communication with control unit

INFORMATION

Communication with the control unit designated in the fault text is/was not possible.

Possible fault causes

- ♦ The power supply for the designated control unit is not present or is present only to a limited extent.
- ♦ The CAN line CAN-High and/or CAN-Low is interrupted between the control unit connector of the designated control unit and the first splice behind the control unit.
- ♦ Control unit faulty

Fault effects

- ♦ Cross-vehicle functions are not available
- ♦ Existing substitute/emergency functions of the control units are active
- ♦ Functions in the control units are available only with restrictions
- ♦ Functions in the control units are not available

000003

Diagnostic information - CAN

Fault entry in the specified control unit

INFORMATION

Internal function test completed.

Diagnostic conditions

The control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- Event-controlled CAN message
- Wake-up signal via wake-up line
- Key contact (terminal S)
- Ignition on

Possible fault causes

- ◆ Incorrect control unit coding
- ◆ The peripheral equipment connected to the specified control unit is faulty
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

Fault effects

- ◆ The stated control unit sends only substitute values via the CAN
- ◆ Other control units detect these substitute values and set this fault code

Diagnostic information - CAN

Fault entry in the specified control unit

INFORMATION

Internal function test completed.

Diagnostic conditions

The control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- Event-controlled CAN message
- Wake-up signal via wake-up line
- Key contact (terminal S)
- Ignition on

Possible fault causes

- ♦ Incorrect control unit coding
- ♦ The peripheral equipment connected to the specified control unit is faulty
- ♦ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

Fault effects

- ♦ The stated control unit sends only substitute values via the CAN
- ♦ Other control units detect these substitute values and set this fault code

Diagnostic information - headlight control unit (central)

Compression angle sensor (only without level control unit (PASM))

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

NOTE

- ◆ On vehicles with level control/PASM, the signal for the vehicle position is generated via the leveling system/PASM control unit. This has four angle sensors for compression.
- ◆ The signal values are then sent to the headlight control unit (central) via CAN
- ◆ Calibration takes place in the **leveling system control unit (PASM)**
- ◆ Calibration takes place in the headlight control unit (central) only without **level control unit (PASM)**

Diagnostic conditions

- ◆ Ignition on

Fault setting condition(s)

- ◆ **Angle sensor** calibration incorrect or not carried out
- ◆ **>Commissioning<** function under Maintenance/repairs not performed

Fault effects

- ◆ No function of headlight beam adjustment

Diagnosis information - Headlight control unit (central)

Supply voltage

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ◆ Function not available or available only to a limited extent

Diagnosis information - Headlight control unit (central)

Supply voltage

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

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Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

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Diagnosis information - Headlight control unit (central)

Supply voltage

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Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ◆ Function not available or available only to a limited extent

Diagnosis information - Headlight control unit (central)

Supply voltage

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The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ◆ Function not available or available only to a limited extent

Diagnostic information - headlight control unit (central)

Compression angle sensor (only without level control unit (PASM))

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

NOTE

- ◆ On vehicles with level control/PASM, the signal for the vehicle position is generated via the leveling system/PASM control unit. This has four angle sensors for compression.
- ◆ The signal values are then sent to the headlight control unit (central) via CAN
- ◆ Calibration takes place in the **leveling system control unit (PASM)**
- ◆ Calibration takes place in the headlight control unit (central) only without **level control unit (PASM)**

Diagnostic conditions

- ◆ Ignition on

Fault setting condition(s)

- ◆ **Angle sensor** calibration incorrect or not carried out
- ◆ **>Commissioning<** function under Maintenance/repairs not performed

Fault effects

- ◆ No function of headlight beam adjustment

Diagnostic information - Headlight control unit (central)

Power supply to HBA angle sensor (not leveling system)

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

NOTE

- ◆ The lights should always be switched off before removing and installing the headlights.
- ◆ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

INFORMATION

- ◆ The angle sensor (compression) is also referred to as the vehicle level sensor.
- ◆ The power supply (5 V) for the angle sensors is supplied via the **Headlight control unit (central)**.
- ◆ HBA (headlight beam adjustment) is also referred to as AHBA (automatic headlight beam adjustment)
- ◆ In the case of the leveling system, the signal comes from the leveling system control unit via CAN

Diagnostic conditions

- **Headlight control unit (central)** internal function test
- Ignition on

Possible fault causes

- ◆ Supply voltage line connection faulty, contacts faulty
- ◆ Ground supply line connection faulty, contacts faulty
- ◆ **Headlight control unit (central)** coded incorrectly
- ◆ Line connection in headlight housing
- ◆ **Headlight control unit (central)**

Diagnostic information - Headlight control unit (central)

Power supply to HBA angle sensor (not leveling system)

CAUTION

Danger of eye injuries due to bright light!

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NOTE

- ◆ The lights should always be switched off before removing and installing the headlights.
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- ◆ In the case of the leveling system, the signal comes from the leveling system control unit via CAN

Diagnostic conditions

- **Headlight control unit (central)** internal function test
- Ignition on

Possible fault causes

- ◆ Supply voltage line connection faulty, contacts faulty
- ◆ Ground supply line connection faulty, contacts faulty
- ◆ **Headlight control unit (central)** coded incorrectly
- ◆ Line connection in headlight housing
- ◆ **Headlight control unit (central)**

Diagnostic information - Headlight control unit (central)

Compression angle sensor (only without **Level-control control unit (PASM)**)

CAUTION

Danger of eye injuries due to bright light!

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INFORMATION

- ◆ In vehicles with level control/PASM, the signal for the vehicle position is generated via the level control/PASM control unit. This has four angle sensors for compression.
- ◆ The signal values are then sent to the headlights via CAN
- ◆ Calibration takes place in the **Level-control control unit (PASM)**
- ◆ Only without **Level-control control unit (PASM)** does Calibration take place in the **Headlight control unit (central)**

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Angle sensor** Calibration incorrect or not carried out
- **Angle sensor Power supply**
- **Angle sensor** Signal lead Faulty
- **Angle sensor** Faulty

Fault effects

- ◆ Headlight beam adjustment Function not available or available only to a limited extent

Diagnostic information - Headlight control unit (central)

Compression angle sensor (only without **Level-control control unit (PASM)**)

CAUTION

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Diagnostic conditions

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- Ignition on

Fault setting condition(s)

- **Angle sensor** Calibration incorrect or not carried out
- **Angle sensor Power supply**
- **Angle sensor** Signal lead Faulty
- **Angle sensor** Faulty

Fault effects

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Diagnostic information - Headlight control unit (central)

Compression angle sensor (only without **Level-control control unit (PASM)**)

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- **Angle sensor Power supply**
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Compression angle sensor (only without **Level-control control unit (PASM)**)

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- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Angle sensor** Calibration incorrect or not carried out
- **Angle sensor Power supply**
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Fault effects

- ◆ Headlight beam adjustment Function not available or available only to a limited extent

Diagnostic information - Headlight control unit (central)

Compression angle sensor (only without **Level-control control unit (PASM)**)

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Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Angle sensor** Calibration incorrect or not carried out
- **Angle sensor Power supply**
- **Angle sensor** Signal lead Faulty
- **Angle sensor** Faulty

Fault effects

- ◆ Headlight beam adjustment Function not available or available only to a limited extent

Diagnostic information - Headlight control unit (central)

Compression angle sensor (only without **Level-control control unit (PASM)**)

CAUTION

Danger of eye injuries due to bright light!

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INFORMATION

- ◆ In vehicles with level control/PASM, the signal for the vehicle position is generated via the level control/PASM control unit. This has four angle sensors for compression.
- ◆ The signal values are then sent to the headlights via CAN
- ◆ Calibration takes place in the **Level-control control unit (PASM)**
- ◆ Only without **Level-control control unit (PASM)** does Calibration take place in the **Headlight control unit (central)**

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Angle sensor** Calibration incorrect or not carried out
- **Angle sensor Power supply**
- **Angle sensor** Signal lead Faulty
- **Angle sensor** Faulty

Fault effects

- ◆ Headlight beam adjustment Function not available or available only to a limited extent

Diagnosis information - Lamp adjustment control unit

Supply voltage

⚠ CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ◆ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnosis information - Lamp adjustment control unit

Supply voltage

⚠ CAUTION

Danger of eye injuries due to bright light!

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Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

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- Ignition on

Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ◆ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnosis information - Headlight control unit (central)

Control module , not coded

Control module , incorrectly coded

NOTE

- ♦ The lights should always be switched off before removing and installing the headlights.
- ♦ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Headlight control unit (central) not coded
- ♦ Headlight control unit (central) incorrectly coded
- ♦ Headlight control unit (central) Faulty

Diagnostic information - CAN

Fault entry in the specified control unit

INFORMATION

Internal function test completed.

Diagnostic conditions

The control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- Event-controlled CAN message
- Wake-up signal via wake-up line
- Key contact (terminal S)
- Ignition on

Possible fault causes

- ♦ Incorrect control unit coding
- ♦ The peripheral equipment connected to the specified control unit is faulty
- ♦ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

Fault effects

- ♦ The stated control unit sends only substitute values via the CAN
- ♦ Other control units detect these substitute values and set this fault code

Diagnosis information - Lamp adjustment control unit

LIN bus

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- **Servo motor** no communication

Possible fault causes

- ◆ Unjustified entry due to software error in some cases; this can be deleted with dipped beam ON
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- ◆ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnosis information - Lamp adjustment control unit

LIN bus

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

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Diagnosis information - Lamp adjustment control unit

LIN bus

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- ◆ Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- ◆ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnosis information - Lamp adjustment control unit

Supply voltage

⚠ CAUTION

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Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ◆ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnosis information - Lamp adjustment control unit

Supply voltage

⚠ CAUTION

Danger of eye injuries due to bright light!

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Fault setting condition(s)

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Supply voltage

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Diagnosis information - Lamp adjustment control unit

Supply voltage

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Diagnosis information - Lamp adjustment control unit

Supply voltage

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- Ignition on

Fault setting condition(s)

- Undervoltage (Voltage < 8 V)
- Overvoltage (Voltage > 16 V)

Fault effects

- ♦ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnostic information - Lamp adjustment control unit

Servo motor / Adjusting motor

CAUTION

Danger of eye injuries due to bright light!

- ♦ Do not look directly into the light beam.

NOTE

- ♦ The lights should always be switched off before removing and installing the headlights.
- ♦ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

The control unit tests internal functional sequences when the ignition is switched on (terminal 15).

All diagnosable connections are checked and any faults are stored.

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Headlights switched on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Line connection or contacts to **Servo motor** faulty
- ♦ Power or ground supply faulty
- ♦ **Servo motor** overheated
- ♦ **Servo motor** mechanical blocked
- ♦ **Servo motor** Faulty
- ♦ **Lamp adjustment control unit** incorrectly coded

Diagnostic information - Lamp adjustment control unit

Servo motor / Adjusting motor

CAUTION

Danger of eye injuries due to bright light!

- ♦ Do not look directly into the light beam.

NOTE

- ♦ The lights should always be switched off before removing and installing the headlights.
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Diagnostic conditions

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All diagnosable connections are checked and any faults are stored.

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Headlights switched on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Line connection or contacts to **Servo motor** faulty
- ♦ Power or ground supply faulty
- ♦ **Servo motor** overheated
- ♦ **Servo motor** mechanical blocked
- ♦ **Servo motor** Faulty
- ♦ **Lamp adjustment control unit** incorrectly coded

Diagnostic information - Lamp adjustment control unit

Servo motor / Adjusting motor

CAUTION

Danger of eye injuries due to bright light!

- ◆ Do not look directly into the light beam.

NOTE

- ◆ The lights should always be switched off before removing and installing the headlights.
- ◆ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

The control unit tests internal functional sequences when the ignition is switched on (terminal 15).

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